CENG463 - Machine Learning

Predicting and Classifying Terrorism Incidents using Machine Learning

Presented by Faruk Kaplan, Ali Emre Pamuk, Mert Altekin

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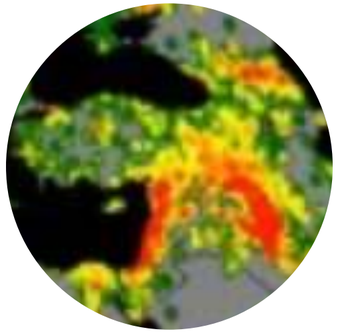
Problem Definiton

- Unpredictability and Complexity: Terrorism poses challenges in predicting attack success and identifying activity patterns.
- Data Processing Limitations: Traditional methods struggle with analyzing large datasets on location, weapon types, and target demographics, resulting in inefficient resource use and delays.
- Proactive Response Deficit: Governments and agencies face difficulties in implementing preventive counterterrorism measures, often reacting instead of preventing attacks.

Problem Definiton



- Consequences and Emerging Threats: Increased casualties, infrastructure damage, ineffective policies, and unconventional attack methods complicate counterterrorism efforts.
- Need for Advanced Tools: A lack of specific risk insights hinders targeted training and preparedness, highlighting the necessity for advanced analytical and predictive tools.

To address these gaps in counterterrorism, we developed a model that performs:

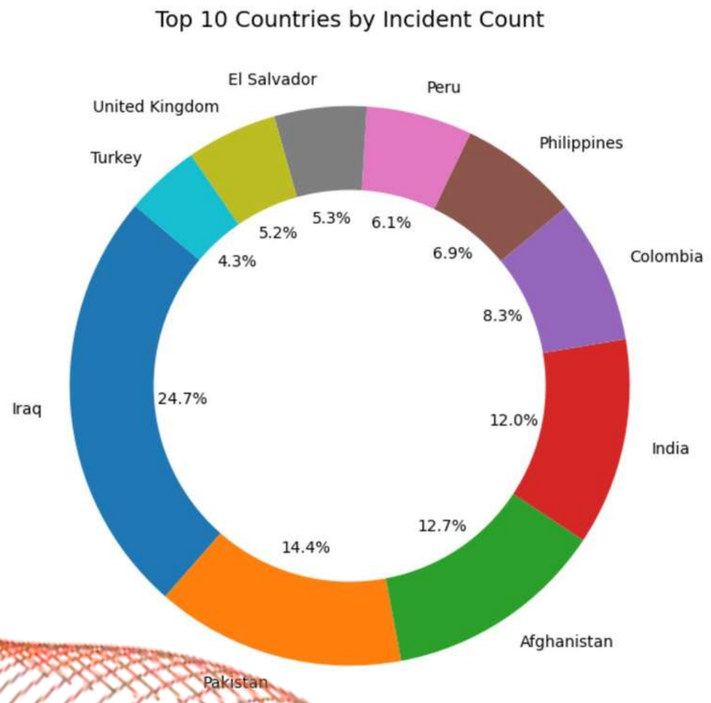
- Classify the type of potential terrorist attacks based on factors such as location, weapon type, target etc..
- Predict of whether will terror incident succeed or not

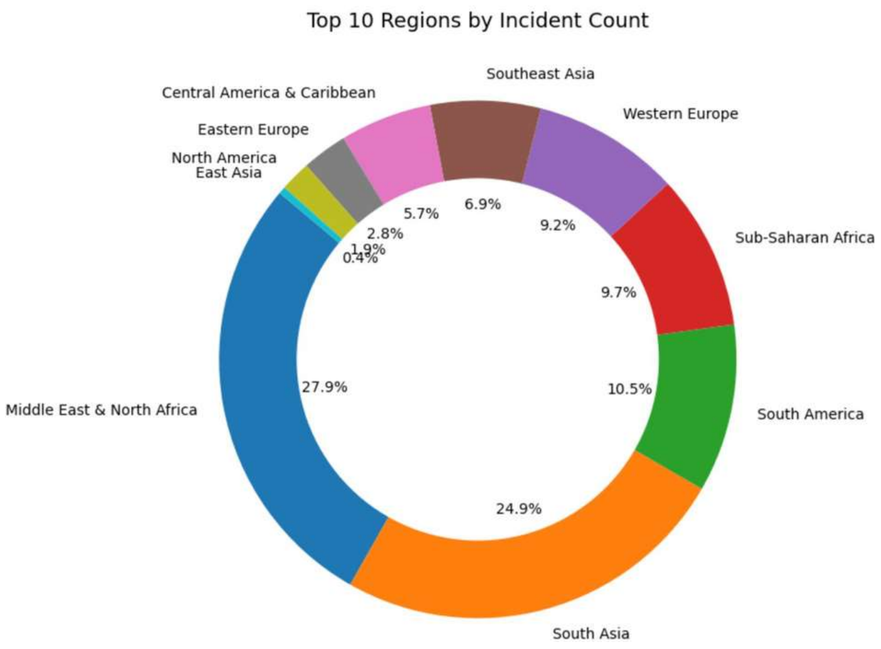
Data Collection and Exploration

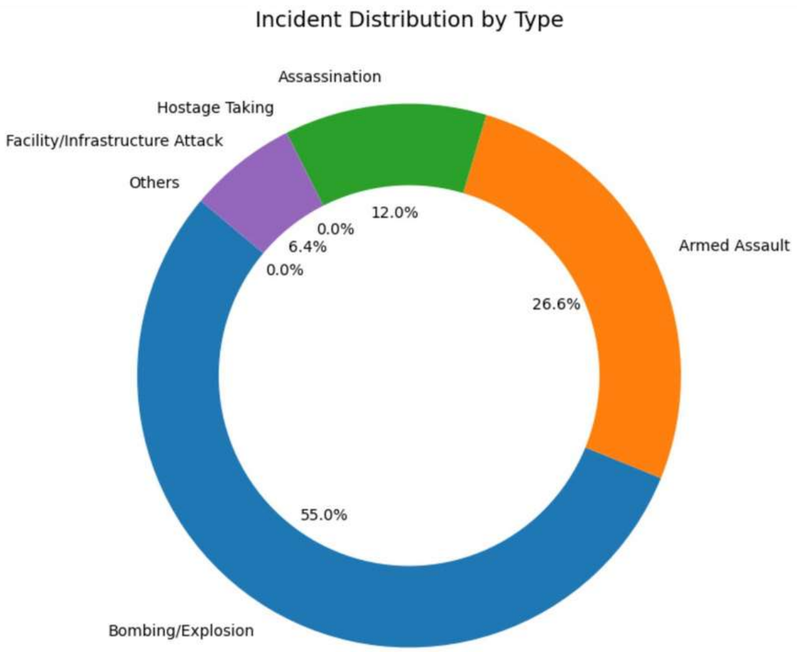
For dataset, we will use global terrorism database

- The Global Terrorism Database (GTD) is an open-source database including information on terrorist attacks around the world from 1970 to 2017.
- The GTD includes systematic data on domestic as well as international terrorist incidents that have occurred during this time period and now includes more than 180,000 attacks.
- The database is maintained by researchers at the National Consortium for the Study of Terrorism and Responses to Terrorism (START), headquartered at the University of Maryland.

Data Collection and Exploration







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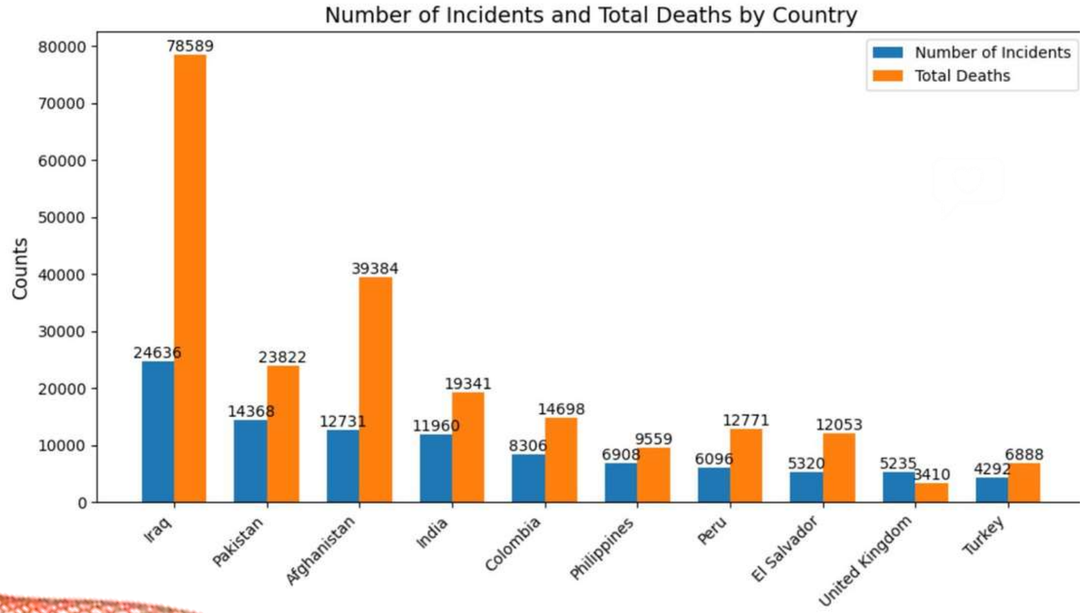

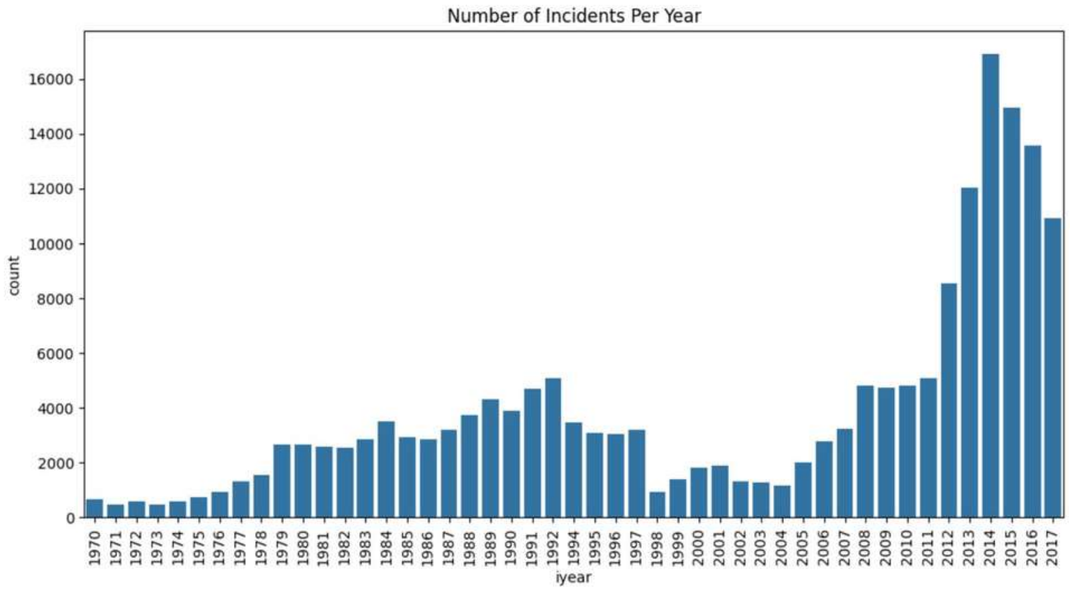

Data Preprocessing

Data Cleaning	Feature Engineering
• Dropping duplicates • Dropping columns that contains mostly NaN values • Removes rows that contains NaN values in critical columns	• Matching the success criteria • Encoding categorical features • Scaling numerical features

Algorithms

	Prediction Model	Classification Model
Logistic Regression	✓	✓
Support Vector Machines (SVM)		✓
Bernoulli Naive Bayes		✓
K Nearest Neighbors (kNN)		✓

Motivation

Logistic Regression in Prediction

- Logistic regression is inherently designed for prediction tasks, which aligns well with the problem of predicting whether a terrorism incident will succeed (1) or fail (0).
- If the features and target variable have a roughly linear relationship in log-odds, logistic regression is an ideal choice as it efficiently models this relationship.
- Logistic regression is computationally efficient.
- Logistic regression can handle imbalanced datasets effectively, ensuring balanced predictions as reflected in the confusion matrix.
- Logistic regression is often used as a strong baseline model to evaluate the feasibility of more complex models. Given its solid performance here, it may be sufficient without requiring more complex approaches.,

Motivation

Classification

Logistic Regression	• Key Characteristics: Logistic regression is simple, interpretable, and computationally efficient, performing well with a linear relationship between features and the log-odds of the target variable. • Interpretability: Enables straightforward understanding of feature contributions, making it ideal for transparent predictions.
Support Vector Machines (SVM)	• Strengths of SVMs: Particularly effective in high-dimensional spaces and for well-separated classes, SVMs excel in complex classification tasks by maximizing the margin between classes. • Versatility: Handles both linear and non-linear relationships, making it adaptable to diverse datasets.

Motivation

Classification

Bernoulli Naive Bayes	• Effective for Binary/Categorical Features: Bernoulli Naive Bayes excels with binary or categorical data, assuming independent impacts of feature presence or absence on classification. • Use Cases: Ideal for rapid prototyping and scenarios prioritizing interpretability and speed.
K Nearest Neighbors (kNN)	• Algorithm Characteristics: kNN is a non-parametric, instance-based algorithm that predicts based on the similarity of data points to their nearest neighbors. • Strengths and Requirements: Intuitive and flexible, kNN benefits from proper feature scaling and optimal parameter tuning (e.g., selecting the best k).

Experimental Scheme

Data Preprocessing

- Removed duplicates and missing values.
- Excluded rows with unknown attack types, months, or days.
- Encoded categorical variables and scaled numerical variables

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Model Training and Evaluation

- Split dataset into 80% training and 20% testing sets.
- Evaluation metrics include:
- Accuracy, precision, recall, f1-score
- Confusion Matrix to analyze true/false positives and negatives.

Results

Logistic Regression for Prediction

Accuracy

The model achieved an accuracy of 85%, indicating a good overall performance in classification tasks.

Missclassifications

The confusion matrix reveals that 1,218 samples from Class 1 were incorrectly predicted, and 1,059 samples from Class 0 were incorrectly.

	precision	recall	f1-score	support
0	0.83	0.85	0.84	7048
1	0.87	0.85	0.86	8114
accuracy			0.85	15162
macro avg	0.85	0.85	0.85	15162
weighted avg	0.85	0.85	0.85	15162

```
Confusion Matrix:
 [[5991 1057]
 [1215 6899]]
```

Results

Logistic Regression for Classification

Accuracy

The model achieved an accuracy of 82.8%, indicating strong overall performance.

Missclassifications

Misclassifications are more frequent for minority classes, showing room for improvement in handling class imbalance

```
Test Accuracy: 0.828122938926263
Classification Report:
              precision    recall  f1-score   support

           1       0.64      0.48      0.55      1629
           2       0.74      0.83      0.79      4472
           3       0.93      0.98      0.95      7075
           4       0.48      0.20      0.29        49
           5       0.69      0.30      0.42       121
           6       0.53      0.22      0.31       646
           7       0.84      0.80      0.82      1069
           8       0.71      0.52      0.60       101

    accuracy                           0.83     15162
   macro avg       0.69      0.54      0.59     15162
weighted avg       0.81      0.83      0.82     15162

Confusion Matrix:
[[ 780  553  247    1    2   32   12    2]
 [ 321 3723  232    6    8   66  112    4]
 [  18   77 6953    2    3    0   18    4]
 [   1   21    9   10    1    5    1    1]
 [  12   44   21    1   36    3    3    1]
 [  67  391   29    0    2  143   11    3]
 [  12  171    7    1    0   13  858    7]
 [  12   25    1    0    0    8    2   53]]
```

Results

SVM for Classification

Accuracy

SVM achieved a slightly higher accuracy (83.2%) compared to Logistic Regression.

Confusion Matrix

The confusion matrix indicates fewer false negatives for majority classes, which contributes to the higher recall values overall.

```
Test Accuracy: 0.8317504287033373
Classification Report:
              precision    recall  f1-score   support

           1       0.63      0.50      0.56      1629
           2       0.75      0.84      0.79      4472
           3       0.93      0.98      0.95      7075
           4       0.59      0.20      0.30        49
           5       0.83      0.28      0.42       121
           6       0.62      0.21      0.31       646
           7       0.85      0.83      0.84      1069
           8       0.74      0.42      0.53       101

    accuracy                           0.83     15162
   macro avg       0.74      0.53      0.59     15162
weighted avg       0.82      0.83      0.82     15162

Confusion Matrix:
[[ 808  544  246    1    1   15   11    3]
 [ 331 3738  239    3    4   46  108    3]
 [  19   77 6958    1    1    0   17    2]
 [   1   21    9   10    0    3    4    1]
 [   7   49   22    1   34    2    5    1]
 [  76  391   30    0    1  136   12    0]
 [  16  153    2    1    0    7  885    5]
 [  19   27    0    0    0    9    4   42]]
```

Results

Bernoulli Naive Bayes for Classification

Accuracy

The accuracy of 80.0% is lower than that of Logistic Regression and SVM, suggesting BNB is less effective on this dataset.

Missclassification

A high number of misclassifications are observed for Classes 4 and 5, reinforcing the interpretation of poor minority class performance.

```
Test Accuracy: 0.8010156971375808
Classification Report:
              precision    recall  f1-score   support

           1       0.56      0.48      0.52      1629
           2       0.75      0.75      0.75      4472
           3       0.93      0.98      0.95      7075
           4       1.00      0.00      0.00        49
           5       0.91      0.08      0.15       121
           6       0.38      0.38      0.38       646
           7       0.73      0.79      0.76      1069
           8       0.62      0.20      0.30       101

    accuracy                           0.80     15162
   macro avg       0.74      0.46      0.48     15162
weighted avg       0.80      0.80      0.79     15162

Confusion Matrix:
[[ 787  482  252    0    0   79   29    0]
 [ 432 3335  229    0    0  262  205    9]
 [  43  101 6903    0    1    1   26    0]
 [   5   23    9    0    0    7    5    0]
 [  30   40   22    0   10    4   15    0]
 [  64  299   27    0    0  244   12    0]
 [  19  167    8    0    0   26  846    3]
 [  15   29    5    0    0   17   15   20]]
```

Results

kNN for Classification

Accuracy

The model achieved an accuracy of 80.8%, slightly better than BNB but lower than Logistic Regression and SVM.

Missclassification

Errors are more dispersed across minority classes, indicating challenges in maintaining class balance during predictions.

```
Test Accuracy: 0.808534494130062
Classification Report:
              precision    recall  f1-score   support

           1       0.56      0.52      0.54      1629
           2       0.74      0.81      0.77      4472
           3       0.91      0.96      0.94      7075
           4       0.67      0.08      0.15        49
           5       0.65      0.26      0.37       121
           6       0.52      0.22      0.31       646
           7       0.84      0.72      0.77      1069
           8       0.90      0.35      0.50       101

    accuracy                           0.81     15162
   macro avg       0.72      0.49      0.54     15162
weighted avg       0.80      0.81      0.80     15162

Confusion Matrix:
[[ 847  506  235    0    3   23   13    2]
 [ 427 3636  235    0    7   88   79    0]
 [  90  164 6798    1    1    0   21    0]
 [   2   20   13    4    2    3    5    0]
 [  14   40   28    0   31    1    6    1]
 [  79  364   47    0    2  141   13    0]
 [  33  179   76    1    2   10  767    1]
 [  13   24   15    0    0    7    7   35]]
```

Trends and Patterns

Prediction	• Balanced Performance: The model demonstrates consistent performance across classes, with similar precision, recall, and F1-scores for Classes 0 and 1, indicating no significant class bias. • Temporal Trends: A sharp rise in incidents post-2000, peaking in recent years, highlights the importance of focusing on high-frequency periods for improving predictive accuracy.
Classification	• Performance on Majority vs. Minority Classes: Majority class (3, Bombing) consistently achieves high precision, recall, and F1-scores across all models, while minority classes (e.g., 4, 5, 8) show significant performance decline, especially with BNB and kNN. • Model Comparison: SVM slightly outperforms Logistic Regression in accuracy and F1-scores for key classes, emerging as the best overall model, while BNB shows the lowest accuracy and poorest performance, making it unsuitable for the dataset. • Balanced Performance: Logistic Regression effectively balances performance across classes compared to kNN and BNB, maintaining high accuracy.

Future Predictions

```
The realistic random attack is predicted to be successful.
Probability of success: 0.94
Realistic random attack details:    iyear  imonth  iday  country  region  attacktype1  targtype1  natlty1  \
0   1980       1    17       56      11            2          5       56

   weaptype1  nkill  nwound country_txt          region_txt attacktype1_txt  \
0          5      8       4    Djibouti  Sub-Saharan Africa   Armed Assault

      targtype1_txt natlty1_txt  \
0  Abortion Related    Djibouti

                                                gname weaptype1_txt
0  Front for the Restoration of Unity and Democracy      Firearms
```

```
The realistic random attack is predicted to be unsuccessful.
Probability of success: 0.04
Realistic random attack details:    iyear  imonth  iday  country  region  attacktype1  targtype1  natlty1  \
0   1996       9    23       54       9            7         19       54

   weaptype1  nkill  nwound     country_txt      region_txt  \
0          8      2      31  Czech Republic  Eastern Europe

                  attacktype1_txt   targtype1_txt     natlty1_txt  \
0  Facility/Infrastructure Attack  Transportation  Czech Republic

                              gname weaptype1_txt
0  Revolutionary Cells Network (SRN)    Incendiary
```

Using the most recent year's data distribution, the model simulated potential attack outcomes

Improvements

Challenges Identified	Proposed Enhancements
• Data imbalance for certain attack types (e.g., rare categories like hijackings). • Limited feature set, as political or economic factors were not included.	• Incorporate external datasets with socioeconomic and geopolitical indicators. • Experiment with ensemble methods like XGBoost to improve accuracy. • Develop a feature selection process to identify the most predictive features.

Conclusion

- Successfully predicted and classified terrorism incidents using machine learning.
- Highlighted the importance of data-driven insights for global security and policymaking.
- Future improvements can expand the model's applicability and accuracy.